

Anchor/clipping Calibration Results

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Outline

- Calibration experiments Delta method on validation data using the W4A4 setting where the Q,K and V matrices are left unquantized
- Experimental Results for the above calibration and setting
- Results when using QuaRot with K and V quantized, and using DuQuant with Q,K and V quantized.

Anchor/Clipping Calibration Setup

- 64 prompts are used from each of these three benchmarks:
 - GSM8K_val
 - MATH_val
 - MBPP_val
- These validation datasets do not contain any prompt from the test benchmarks to avoid data leakage

DeltaDuQuant calibration on LLaDa-8B-Base Model

Table 1: Anchor scheduling results for DuQuant-64ex base model. Configurations are ranked by average benchmark score within each stage.

Stage	Anchor Type	# Anchors	Avg. $\uparrow$	GSM8K	MATH	MBPP
Stage 1	uniform	32	51.04	68.75	48.44	35.94
	online hybrid	32	50.00	70.31	45.31	34.38
	offline	16	49.48	67.19	43.75	37.50
	uniform	16	48.96	71.88	43.75	31.25
	online hybrid	2	48.96	68.75	45.31	32.81
	online hybrid	8	48.44	65.63	45.31	34.38
	uniform	2	47.92	68.75	40.63	34.38
	offline	8	47.92	64.06	43.75	35.94
	online hybrid	4	47.92	67.19	43.75	32.81
	online hybrid	16	47.92	65.63	43.75	34.38
	offline	32	47.40	67.19	42.19	32.81
	uniform	8	46.88	68.75	37.50	34.38
	uniform	1	46.35	65.63	39.06	34.38
	exponential	4	46.35	59.38	45.31	34.38
	offline	4	45.83	64.06	37.50	35.94
	uniform	4	45.31	64.06	37.50	34.38
	offline	2	44.27	59.38	39.06	34.38
Stage 2	uniform + clip 0.99	32	51.56	70.31	48.44	35.94
	uniform + clip 0.999	32	48.44	71.88	37.50	35.94
	uniform + clip 0.95	32	47.92	68.75	37.50	37.50
	uniform + clip 0.9	32	47.40	64.06	45.31	32.81

DeltaQuarot calibration on LLaDa-8B-Base Model

Table 2: Anchor scheduling results for QuaRot-64ex base model. Configurations are ranked by average benchmark score within each stage.

Stage	Anchor Type	# Anchors	Avg. $\uparrow$	GSM8K	MATH	MBPP
Stage 1	uniform	32	55.73	73.44	50.00	43.75
	uniform	16	52.08	70.31	45.31	40.63
	offline	32	52.08	73.44	40.63	42.19
	uniform	8	51.56	78.13	37.50	39.06
	offline	4	51.56	76.56	40.63	37.50
	online hybrid	2	51.56	75.00	40.63	39.06
	online hybrid	16	51.56	73.44	42.19	39.06
	offline	16	51.04	71.88	40.63	40.63
	uniform	1	50.52	75.00	37.50	39.06
	uniform	4	50.52	75.00	39.06	37.50
	exponential	4	50.52	59.38	48.44	43.75
	online hybrid	4	50.52	71.88	42.19	37.50
	online hybrid	8	50.52	71.88	42.19	37.50
	offline	2	50.00	73.44	39.06	37.50
	online hybrid	32	49.48	67.19	39.06	42.19
	offline	8	48.96	68.75	37.50	40.63
	uniform	2	48.44	68.75	37.50	39.06
Stage 2	uniform + clip 0.999	32	49.48	64.06	39.06	45.31
	uniform + clip 0.99	32	48.96	68.75	39.06	39.06
	uniform + clip 0.95	32	47.92	60.94	40.63	42.19
	uniform + clip 0.9	32	47.40	64.06	39.06	39.06

DeltaDuQuant calibration on LLaDa-8B-Instruct Model

Table 3: Anchor scheduling results for DuQuant-64ex instruct model. Configurations are ranked by average benchmark score within each stage.

Stage	Anchor Type	# Anchors	Avg. $\uparrow$	GSM8K	MATH	MBPP
Stage 1	uniform	32	58.85	85.94	68.75	21.88
	exponential	4	57.81	84.38	65.63	23.44
	online hybrid	16	57.81	84.38	70.31	18.75
	uniform	16	57.29	78.13	68.75	25.00
	online hybrid	8	57.29	84.38	71.88	15.63
	offline	4	56.25	78.13	67.19	23.44
	offline	8	56.25	82.81	65.63	20.31
	offline	32	56.25	79.69	67.19	21.88
	online hybrid	2	55.73	82.81	60.94	23.44
	uniform	8	55.21	81.25	68.75	15.63
	offline	2	55.21	79.69	67.19	18.75
	online hybrid	32	55.21	79.69	64.06	21.88
	uniform	2	54.69	82.81	65.63	15.63
	offline	16	54.69	78.13	70.31	15.63
	online hybrid	4	54.69	79.69	67.19	17.19
	uniform	1	54.17	79.69	67.19	15.63
	uniform	4	53.65	78.13	64.06	18.75
Stage 2	uniform + clip 0.95	32	57.81	85.94	65.63	21.88
	uniform + clip 0.9	32	57.29	79.69	70.31	21.88
	uniform + clip 0.99	32	56.25	82.81	64.06	21.88
	uniform + clip 0.999	32	54.17	81.25	62.50	18.75

DeltaQuaRot calibration on LLaDa-8B-Instruct Model

Table 4: Anchor scheduling results for QuaRot-64ex instruct model. Configurations are ranked by average benchmark score within each stage.

Stage	Anchor Type	# Anchors	Avg. $\uparrow$	GSM8K	MATH	MBPP
Stage 1	offline	16	59.38	81.25	65.63	31.25
	offline	32	56.25	75.00	64.06	29.69
	uniform	16	55.73	78.13	60.94	28.13
	uniform	32	55.73	76.56	62.50	28.13
	offline	8	55.73	76.56	62.50	28.13
	online hybrid	4	55.73	78.13	60.94	28.13
	online hybrid	32	55.21	78.13	59.38	28.13
	uniform	1	54.69	81.25	57.81	25.00
	offline	2	54.69	79.69	59.38	25.00
	online hybrid	8	54.69	71.88	64.06	28.13
	online hybrid	16	54.69	75.00	60.94	28.13
	uniform	2	54.17	76.56	59.38	26.56
	exponential	4	54.17	73.44	60.94	28.13
	uniform	4	53.65	76.56	59.38	25.00
	offline	4	53.65	71.88	62.50	26.56
	uniform	8	53.13	70.31	62.50	26.56
	online hybrid	2	52.08	70.31	59.38	26.56
Stage 2	offline + clip 0.9	16	56.77	75.00	70.31	25.00
	offline + clip 0.95	16	55.73	75.00	67.19	25.00
	offline + clip 0.99	16	55.21	73.44	64.06	28.13
	offline + clip 0.999	16	55.21	78.13	59.38	28.13

Evaluation on LLaDa Instruct Model

Table 7:

LLaDA-8B-Instruct accuracy: our reproduction via Fast-dLLM (prefix cache, real FlashAttention) vs. FreeAct’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	MBPP	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base	43.29	77.10	38.4	27.80	46.65	—
	W4A4	QuaRot	32.32	68.23	28.4	26.40	38.84	16.74
		DeltaQuaRot	34.15	72.02	31.0	26.20	40.84	12.44
		DeltaQuaRot (offline, 16 anc.)	34.76	71.95	33.6	28.60	42.23	9.48
		DuQuant	36.59	73.84	36.4	23.20	42.51	8.88
		DeltaDuQuant	42.07	76.04	36.0	—	—	—
		DeltaDuQuant (uniform, 32 anc.)	40.24	75.21	36.4	—	—	—

Comparison with FreeAct

Table 5:

LLaDA-8B-Instruct accuracy: our reproduction via Fast-dLLM (prefix cache, real FlashAttention) vs. FreeAct’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base	43.29	77.10	38.4	52.93	–
	W4A4	QuaRot	32.32	68.23	28.4	42.98	18.80
		DeltaQuaRot	34.15	72.02	31.0	45.72	13.62
		DeltaQuaRot (offline, 16 anc.)	34.76	71.95	33.6	46.77	11.64
		DuQuant	36.59	73.84	36.4	48.94	7.54
		DeltaDuQuant	42.07	76.04	36.0	51.37	2.95
		DeltaDuQuant (uniform, 32 anc.)	40.24	75.21	36.4	50.62	4.37
		FreeAct	42.68	75.74	34.60	51.01	3.63

Evaluation On LLaDa Base Model

Table 8:
LLaDA-8B accuracy: our reproduction via Fast-dLLM vs. DLLMQuant’s and STaR-Quant’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	MBPP	Avg.	$\Delta\%$
LLaDA-8B	FP16	Base	32.93	66.64	33.4	39.40	43.09	–
	W4A4	QuaRot	26.22	61.71	24.0	—	—	—
		DeltaQuaRot	29.27	61.56	29.0	36.20	39.01	9.48
		DeltaQuaRot (uniform, 32 anc.)	31.10	60.27	29.4	36.60	39.34	8.70
		DuQuant	25.00	57.16	25.8	33.40	35.34	17.99
		DeltaDuQuant	29.88	57.85	27.6	36.20	37.88	12.09
		DeltaDuQuant (uniform, 32 anc., clip 0.99)	25.00	59.97	30.8	36.40	38.04	11.72

Comparison with DLLMQuant and STaR-Quant

Table 6:

LLaDA-8B accuracy: our reproduction via Fast-dLLM vs. DLLMQuant’s and STaR-Quant’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	Avg.	$\Delta\%$
LLaDA-8B	BF16	Base	32.93	66.64	49.78	–
	W4A4	QuaRot	26.22	61.71	43.97	11.69
		DeltaQuaRot	29.27	61.56	45.41	8.78
		DeltaQuaRot (uniform, 32 anc.)	31.10	60.27	45.69	8.23
		DuQuant	25.00	57.16	41.08	17.49
		DeltaDuQuant	29.88	57.85	43.87	11.89
		DeltaDuQuant (uniform, 32 anc., clip 0.99)	25.00	59.97	42.48	14.65
		DLLMQuant ⁺⁺	28.92	56.25	42.59	14.46
		STaR-Quant	35.98	57.29	46.63	6.33

Evaluating QuaRot K4V4 and DuQuant Q4K4V4 On LLaDa Instruct Model

Table 17:

LLaDA-8B-Instruct accuracy restricted to HumanEval, GSM8K, and MATH500, matching the benchmarks reported by FreeAct, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base	43.29	77.10	38.4	52.93	–
	W4A4, K4V4	QuaRot	30.49	68.23	27.6	42.11	20.45
		DeltaQuaRot	34.15	72.02	31.0	45.72	13.62
	W4A4, Q4K4V4	DuQuant	37.20	73.16	35.2	48.52	8.33
		DeltaDuQuant	39.02	73.84	36.8	49.89	5.75
	W4A4 (reported)	FreeAct	42.68	75.74	34.60	51.01	3.63

Evaluating QuaRot K4V4 and DuQuant Q4K4V4 On LLaDa Base Model

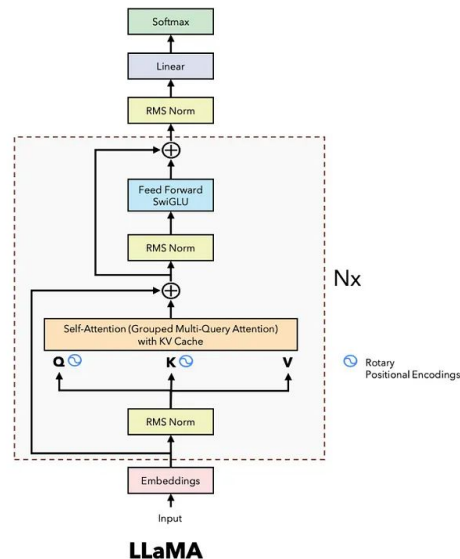
Table 14:
LLaDA-8B accuracy across K4V4 and Q4K4V4 attention scopes, and vs. DLLMQuant’s/STaR-Quant’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	MBPP	Avg.	$\Delta\%$
LLaDA-8B	BF16	Base	32.93	66.64	33.4	39.40	43.09	–
	W4A4, K4V4	QuaRot	23.78	—	23.0	32.00	-	-
		DeltaQuaRot	29.27	—	29.0	36.20	-	-
	W4A4, Q4K4V4	DuQuant	24.39	—	22.2	33.60	-	-
		DeltaDuQuant	22.56	—	26.2	36.40	-	-
	W4A4 (reported)	DLLMQuant ⁺⁺	28.92	56.25	—	—	-	-
		STaR-Quant	35.98	57.29	—	—	-	-

Memory Needed For Caching in LLaDA

- For sequence length of 256 and batch size 1 each input activation will have a shape of $(1, 256, \text{in_features})$. Each output activation will have a size of $(1, 256, \text{out_features})$:

```
LLaDAModelLM(  
  (model): LLaDAModel(  
    (transformer): ModuleDict(  
      (wte): Embedding(126464, 4096)  
      (emb_drop): Dropout(p=0.0, inplace=False)  
      (ln_f): RMSLayerNorm()  
      (blocks): ModuleList(  
        (0-31): 32 x LLaDALlamaBlock(  
          (dropout): Dropout(p=0.0, inplace=False)  
          (act): SiLU()  
          (attn_out): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (ff_out): SINQLinear(in_features=12288, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (rotary_emb): RotaryEmbedding()  
          (attn_norm): RMSLayerNorm()  
          (ff_norm): RMSLayerNorm()  
          (q_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (k_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (v_proj): SINQLinear(in_features=4096, out_features=4096, bias=False, device=cpu, W_q.device=cpu)  
          (ff_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cpu, W_q.device=cpu)  
          (up_proj): SINQLinear(in_features=4096, out_features=12288, bias=False, device=cpu, W_q.device=cpu)  
        )  
      )  
      (ff_out): Linear(in_features=4096, out_features=126464, bias=False)  
    )  
  )  
)
```



Memory Needed For Caching in LLaDA

- For sequence length of 256 and batch size 1 each input activation will have a shape of (1,256,in_features). Each output activation will have a size of (1,256,out_features):

Table 38: Activation memory per LLaDA transformer block (seq_len = 256, batch = 1, fp16/bf16, 2 bytes/element)

Component	Input Mem (bytes)	Output Mem (bytes)	Total (MiB)
q-proj + k-proj + v-proj	6,291,456	6,291,456	12.0
attn_out	2,097,152	2,097,152	4.0
ff_proj + up_proj	4,194,304	12,582,912	16.0
ff_out	6,291,456	2,097,152	8.0
Per-layer total	18,874,368	23,068,672	40.0

Table 39: Total activation memory across all 32 LLaDA transformer blocks

Metric	Value
Per-layer activation memory	40.0 MiB
Number of layers	32
Total activation memory (bytes)	1,342,177,280
Total activation memory	1,280 MiB = 1.25 GiB

Plan for Next Week

- Run missing experiments
- Work on the report